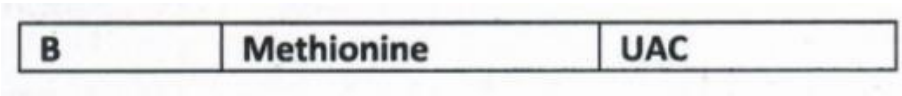
CLASS XII
BIOLOGY (044)
MARKING SCHEME
TERM 1 (2021-22)

<table>
<tr><th>Q.NO.</th><th>ANSWER</th><th>MARKS</th></tr>
<tr><td></td><td>SECTION- A</td><td></td></tr>
<tr><td>1.</td><td>D. 2 thecae, 4 sporangia</td><td>1</td></tr>
<tr><td>2.</td><td>B. 3,3,2
3 in chalazar end 3 in the micropolar end and 2 nuclei in the center.</td><td>1</td></tr>
<tr><td>3.</td><td>B. Free nuclear endosperm</td><td>1</td></tr>
<tr><td>4.</td><td>A. sporopollenin</td><td>1</td></tr>
<tr><td>5.</td><td>B. ii, iii</td><td>1</td></tr>
<tr><td>6.</td><td>A) (i) and (iv)</td><td>1</td></tr>
<tr><td>7.</td><td>C. blastocyst, Fertilized egg, Unfertilized egg</td><td>1</td></tr>
<tr><td>8.</td><td>B. completion of meiosis II</td><td>1</td></tr>
<tr><td>9.</td><td>C. FSH, estrogen, progesterone</td><td>1</td></tr>
<tr><td>10.</td><td>C. small, White, Small, covered with mucilage</td><td>1</td></tr>
<tr><td>11.</td><td>C. Strawberry</td><td>1</td></tr>
<tr><td>12.</td><td>B. 2</td><td>1</td></tr>
<tr><td>13.</td><td>D. i, ii and iv</td><td>1</td></tr>
<tr><td>14.</td><td>D. Glutamic acid is substituted by Valine in β chain at the sixth position</td><td>1</td></tr>
<tr><td>15.</td><td>D. Polygenic and quantitative inheritance</td><td>1</td></tr>
<tr><td>16.</td><td>B. Male 16, Female 32</td><td>1</td></tr>
<tr><td>17.</td><td><table><tr><td></td><td>Rajesh</td><td>Mahesh</td></tr><tr><td>B</td><td>Thalassemia – an autosome linked recessive blood disorder</td><td>Sickle cell anaemia - an autosome linked recessive trait</td></tr></table></td><td>1</td></tr>
<tr><td>18.</td><td>D. (ii) and (iv)</td><td>1</td></tr>
<tr><td>19.</td><td>B. 5′ (upstream) end and 3′ (downstream) end, respectively of the transcription unit</td><td>1</td></tr>
</table>

20.	B. exons appear but introns do not appear in the mature RNA	1
21.	A. lactose is present, and it binds to the repressor	1
22.	B. DNase inhibited transformation	1
23.		1
24.	A. Probes	1
SECTION - B		
25.	C. A is true but R is false	1
26.	D. A is False but R is true	1
27.	C. A is true but R is False.	1
28	C. A is true but R is false	1
29.	A. i and ii	1
30.	A. antipodal, zygote and endosperm	1
31.	A. The flower type which survived is Cleistogamous and it will always exhibit autogamy	1
32.	D. activate smooth muscles	1
33.	C. IUT	1
34.	B. decrease the movement of the sperms	1
35.	C. 500	1
36.	D. 60 Out of 9:3:3:1 = 16 9+3 will be tall. Therefore, 12/16 x 80 = 60.	1
37.	C. 2 Red: 2 Pink	1
38.	D. Aa x aa	1
39.	B. Autosomal recessive	1
40.	B. 50%	1
41.	B. Ab X Ab	1

42.	B. It is a single stranded DNA	1
43.	C. 40,000 bp and 13,600 $x10^{-9}$ m	1
44.	B. A is having 2′-OH group which makes it more reactive and structurally unstable whereas B is having 2′-H group which makes it less reactive and structurally stable	1
45.	D. 0:1:31	1
46.	C. (i) Capping (ii) Polyadenylation (iii) $^{m}G_{ppp}$. (iv) Poly(A).	1
47.	C. Short non-coding repetitive sequence forming large portion of eukaryotic genome	1
48.	C. Children 1 & 3	1
SECTION - C		
49.	C. luteinizing hormone	1
50.	B. Progesterone	1
51.	C. There will be no observed data for Hormone B	1
52.	A. Corpus Luteum	1
53.	D. 280 days	1
54.	B) Subject 2 is pregnant	
55.	B. ii, iii, iv, v	1
56.	C. Affected individual is a female with Down's syndrome	1
57.	D. Deviation from 9:3:3:1 ratio because of linkage of genes	1
58.	C. Translation- Elongation	1
59.	D. (i)- continuous synthesis , (ii)- discontinuous synthesis (iii) 3' end (iv) 5'end	1
60.	C: (i) Promotor Site, (ii) Sigma factor (iii) RNA polymerase	1

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Marking Scheme in lieu of diagram based questions for VI candidates Total Alternative Questions - 20	
Section - A	
2.	C. one meiotic and three mitotic divisions
5.	C. nucellus One meiotic and 3 mitotic divisions.
7.	C. Unfertilized egg/ Fertilized egg/ Blastocyst
10.	B. water
23.	C. i, ii, and iv
Section - B	
29.	C. secretes oxytocin
39.	A. It verifies that DNA is the carrier of genetic information.
44.	D. Hydroxyl
48.	D. 50% bands similar to father and rest similar to mother
Section – C	
	A biology student after studying about the different levels of hormones during the menstrual cycle was comparing 2 subjects (Patients). A table was created after looking at the levels of hormones A and B for Subject 1 and 2. Read the information in the table and answer the questions that follow (Q49 to 54):
49.	C. Luteinizing Hormone
50.	B. Progesterone
51.	C. There will be no observed data for Hormone B
52.	A. Corpus Luteum
53.	D. 280 days
54.	B. Subject 2 is pregnant
56.	C. Due to failure of cytokinesis after telophase stage of cell division
57.	C. Aabb & aaBb
58.	C. to the small subunit; on the large subunit.
59.	A. DNA polymerase can read only in the direction of 3' to 5'
60.	C. When a rho site is reached.

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